

Hypothesis Tests And Confidence Intervals In Multiple Regression

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Chapter Outline

- Hypothesis tests and confidence intervals for one coefficient.
- Joint hypothesis tests on multiple coefficients.
- Other types of hypotheses involving multiple coefficients.
- Confidence sets for multiple coefficients.
- Model specification for multiple regression.

Hypothesis Tests and Confidence Intervals for a Single Coefficient

- Hypothesis tests and confidence intervals for a single coefficient in multiple regression follow the same logic and recipe as for the slope coefficient in a single-regressor model

$$\frac{\hat{\beta}_1 - E[\hat{\beta}_1]}{\sigma_{\hat{\beta}_1}} = \frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1). \quad (1)$$

- Thus hypotheses on β_1 can be tested using the usual *t-statistic*, and confidence intervals are constructed as before.
- The same applies to $\beta_2, \dots, \beta_K$.

Case Study

Class Size and Students' Performance: The California Data

$$\widehat{TestScore} = 698.9 - \frac{2.28}{(10.4)} \times STR, \quad (2)$$

$$R^2 = 0.05, \quad SER = 18.6.$$

$$\widehat{TestScore} = 686 - \frac{1.1}{(8.7)} \times STR, - \frac{0.65}{(0.031)} \times PctEL \quad (3)$$

$$R^2 = 0.426, \quad \bar{R}^2 = 0.424, \quad SER = 14.5$$

- The coefficient on STR in (3) is the effect on $TestScores$ of a unit change in STR , holding constant the percentage of English Learners in the district.
- The coefficient on STR falls considerably but the t -statistic rejects the null hypothesis $\beta_1 = 0$ at $\alpha = 0.05$.

OLS Regression: *STATA* Output

```
reg testscr str pctl, robust;
```

Regression with robust standard errors

Number of obs = 420
F(2, 417) = 223.82
Prob > F = 0.0000
R-squared = 0.4264
Root MSE = 14.464

		Robust				
testscr		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
-----+-----						
str		-1.101296	.4328472	-2.54	0.011	-1.95213 - .2504616
pctl		-.6497768	.0310318	-20.94	0.000	-.710775 - .5887786
_cons		686.0322	8.728224	78.60	0.000	668.8754 703.189

We use **heteroskedasticity-robust standard errors**—for exactly the same reason as in the case of a single regressor.

Tests of Joint Hypotheses

- Let *Expn* be *expenditures per pupil* and consider the population regression model

$$TestScore = \beta_0 + \beta_1 \times STR_i + \beta_2 \times Expn_i + \beta_3 \times PctEL_i + u_i$$

- The null hypothesis that “school resources do not matter”, and the alternative that they do, corresponds to

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \text{ vs. } H_1 : \beta_1 \neq 0, \text{ or } \beta_2 \neq 0 \text{ or both.}$$

- A *joint hypothesis* specifies a value for two or more coefficients, that is, it imposes a restriction on two or more coefficients.
- In general, a joint hypothesis will involve q restrictions. In the our example, $q = 2$ (i.e., $\beta_1 = 0$ and $\beta_2 = 0$).

Tests of Joint Hypotheses

$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \text{ vs. } H_1 : \beta_1 \neq 0, \text{ or } \beta_2 \neq 0 \text{ or both.}$

- A “common sense” idea is to reject if either of the individual *t*-statistics exceeds 1.96 in absolute value.
- But this “one at a time” test is not valid: the resulting test rejects too often under the null hypothesis (i.e., more than 5% of the times when the null hypothesis is true)!
- We can show that the probability of type-1 error of this “one at a time test” is more than 5%.

Tests of Joint Hypotheses

- The *size* of a test is the actual rejection rate under the null hypothesis. The size of the “one at a time” test is not 5%.
- In fact, its size depends on the correlation between t_1 and t_2 , and thus on the correlation between $\hat{\beta}_1$ and $\hat{\beta}_2$.
- Two solutions:
 - ① Use a different critical value in this procedure –not 1.96 (this is the “Bonferroni method” but it is rarely used in practice).
 - ② Use a different test statistic designed to test both β_1 and β_2 at once: the F-statistic (this is common practice).

Tests of Joint Hypotheses

The *F*-statistic

- The *F*-statistic tests all parts of a joint hypothesis at once.
- The formula for the special case of the joint null hypothesis $\beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$ in a regression with two regressors is

$$F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right), \quad (4)$$

- where $\hat{\rho}_{t_1, t_2}$ estimates the correlation between t_1 and t_2 .
- *Rule:* reject when F is large (how large?)

Tests of Joint Hypotheses

The *F*-statistic

- The *F*-statistic is large when t_1 and/or t_2 is large.
- The *F*-statistic corrects (in just the right way) for the correlation between t_1 and t_2 .
- The formula for more than two β 's is nasty unless you use matrix algebra.
- This gives the *F*-statistic a nice large-sample approximate distribution, which is...

Tests of Joint Hypotheses

Large-Sample Distribution of the F-statistic

- Consider the special case in which t_1 and t_2 are independent so $\hat{\rho}_{t_1, t_2} \xrightarrow{P} 0$; in large samples the formula becomes

$$F = \frac{1}{2} \left(\frac{t_1^2 + t_2^2 - 2\hat{\rho}_{t_1, t_2} t_1 t_2}{1 - \hat{\rho}_{t_1, t_2}^2} \right) \approx \frac{1}{2} (t_1^2 + t_2^2). \quad (5)$$

- Under the null, t_1 and t_2 have standard normal distributions that, in this special case, are independent.
- The large-sample distribution of the *F-statistic* is the distribution of the average of two independently distributed squared standard normal random variables.

Tests of Joint Hypotheses

The Chi-Squared Distribution

- The *Chi-squared* distribution with q degrees of freedom (χ_q^2) is defined to be the distribution of the sum of q independent squared standard normal random variables.
- In large samples, the *F-statistic* is distributed as a χ_q^2/q , or also denoted as $F_{q,\infty}$.

q	5% critical value
1	3.84
2	3.00
3	2.60
4	2.37
5	2.21

Tests of Joint Hypotheses

Computing The p -value Using The F -statistic

- The p -value is the tail probability of the χ^2_q/q distribution beyond the F -statistic actually computed.
- Suppose you want to test the joint hypothesis that the population coefficients on STR and expenditures per pupil $expn_stu$ are both zero, against the alternative that at least one of the population coefficients is non-zero.

OLS Regression: STATA Output

```
reg testscr str expn_stu pctel, r;  
Regression with robust standard errors
```

```
Number of obs =      420  
F( 3, 416) = 147.20  
Prob > F      = 0.0000  
R-squared     = 0.4366  
Root MSE     = 14.353
```

		Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
testscr							
str		-.2863992	.4820728	-0.59	0.553	-1.234001	.661203
expn_stu		.0038679	.0015807	2.45	0.015	.0007607	.0069751
pctel		-.6560227	.0317844	-20.64	0.000	-.7185008	-.5935446
_cons		649.5779	15.45834	42.02	0.000	619.1917	679.9641

NOTE

```
test str expn_stu; The test command follows the regression
```

```
(1) str = 0.0 There are q=2 restrictions being tested
```

```
(2) expn_stu = 0.0
```

```
F(2,416) = 5.43 The 5% critical value for q=2 is 3.00
```

```
Prob > F = 0.0047 Stata computes the p-value for you
```

Tests of Joint Hypotheses

Homoskedasticity-only *F-statistic*

- There is a simple formula for the *F-statistic* that holds only under homoskedasticity (so it is not very useful) but which nevertheless might help you understand what the *F-statistic* is doing.
- To compute the *F-statistic* in the homoskedasticity case
 - Run two regressions, one under the null hypothesis (the “restricted” regression) and one under the alternative hypothesis (the “unrestricted” regression).
 - Compare the fit of the regressions –the R^2 ’s– if the “unrestricted” model fits sufficiently better, then reject the null.

Tests of Joint Hypotheses

- The unrestricted regression (that is, under H_1) is

$$TestScore_i = \beta_0 + \beta_1 \times STR_i + \beta_2 \times expn_stu_i + \beta_3 \times PctEL_i + u_i$$

- The restricted population regression (that is, under H_0)

$$TestScore_i = \beta_0 + \beta_3 \times PctEL_i + u_i$$

- The number of restrictions under H_0 is $q = 2$.
- The fit will be better (R^2 will be higher) in the unrestricted regression (why?)
- By how much must the R^2 increase for the coefficients on $Expn$ and $PctEL$ to be judged *jointly* statistically significant?

Tests of Joint Hypotheses

Homoskedasticity-only *F*-statistic

- The homoskedasticity-only formula for the *F*-statistic is

$$F = \frac{(R^2_{unrestricted} - R^2_{restricted}) / q}{(1 - R^2_{unrestricted}) / (n - k_{unrestricted} - 1)}. \quad (6)$$

- The bigger the difference between the restricted and unrestricted R^2 's –the greater the improvement in fit by adding the variables in question– the larger is the *homoskedasticity-only F*-statistic.
- Example: compute the *homoskedasticity-only F*-statistic for our case study.

Example

Restricted Regression:

$$\widehat{TestScore} = 644.7 - 0.671 \times PctEL, \quad R^2_{restricted} = 0.4149 \quad (7)$$

(1.0) (0.032)

Unrestricted regression:

$$\widehat{TestScore} = 649.6 - 0.29 \times STR + 3.87 \times expn_stu - 0.656 \times PctEL,$$

(15.5) (0.48) (1.59) (0.032)

$$R^2_{unrestricted} = 0.4366, k_{unrestricted} = 3, q = 2 \quad (8)$$

$$\text{Thus } F = \frac{(R^2_{unrestricted} - R^2_{restricted})/q}{(1 - R^2_{unrestricted})/(n - k_{unrestricted} - 1)} = \frac{(.4366 - .4149)/2}{(1 - .4366)/(420 - 3 - 1)} = 8.01.$$

Note: Heteroskedasticity-robust $F = 5.43$.

Tests of Joint Hypotheses

Homoskedasticity-only *F*-statistic

- The *homoskedasticity-only F-statistic* rejects the null hypothesis when adding the two variables significantly increases the R^2 of the regression.
- If the errors are homoskedastic, then the *homoskedasticity-only F-statistic* has a large-sample distribution that is χ_q^2/q .
- But if the errors are heteroskedastic, the large-sample distribution of the *homoskedasticity-only F-statistic* is not χ_q^2/q .

Tests of Joint Hypotheses

The F Distribution

- Under the extended OLS assumptions I-V in chapter 5 (and obviously without perfect multicollinearity), the *homoskedasticity-only* F -statistic has the “ $F_{q,n-K-1}$ ” distribution, where q is the number of restrictions and K is the number of regressors under the alternative hypothesis (the unrestricted model).
- The F distribution is to the χ^2/q distribution what the t_{n-1} distribution is to the $N(0, 1)$ distribution.

The F distribution

- The F distribution is tabulated in many places.
- As $n \rightarrow \infty$, the $F_{q,n-k-1}$ distribution converges asymptotically to the χ^2/q distribution:

$$F_{q,\infty} = \chi^2/q$$

- For q not too big and $n \geq 100$, the $F_{q,n-k-1}$ distribution and the χ^2/q distribution are essentially identical.
- Many regression packages compute p -values of F -Statistics using the F distribution.
- You will encounter the F distribution in published empirical work.

Tests of Joint Hypotheses

Summary: The *Homoskedasticity-only F-Statistic* and *F* Distribution

- These are justified only under very strong conditions –not realistic in practice.
- When the sample size is large, you should use the *heteroskedasticity-robust F-statistic*, with distribution χ^2_q/q (that is, $F_{q,\infty}$ critical values).
- For $n \geq 100$, the *F*-distribution essentially is the χ^2_q/q distribution.
- For small n , sometimes researchers use the *F* distribution because it has larger critical values and in this sense is more conservative.

Tests of Joint Hypotheses

Summary: The *Homoskedasticity-only F-Statistic* and *F* Distribution

- The “one at a time” approach of rejecting if either of the *t-statistics* exceeds 1.96 rejects more than 5% of the times under the null (the size exceeds the desired significance level).
- The *heteroskedasticity-robust F-statistic* is built into R; this tests all q restrictions at once.
- For n large, the *F-statistic* is distributed as $\chi_q^2/q (= F_{q,\infty})$. The *homoskedasticity-only F-statistic* can help intuition, but is not valid when there is heteroskedasticity.

Tests of Joint Hypotheses

The “Overall” Regression *F-Statistic*

The “overall” regression *F-statistic* tests the joint hypothesis that *all* slope coefficients are zero. This is

$$H_0 : \beta_1 = \cdots = \beta_K = 0 \text{ vs. } H_1 : \beta_j \neq 0 \text{ for at least one } j.$$

If the sample size is large, the *F-statistic* has an $F_{K,\infty}$ distribution under the null hypothesis.

Example

Dependent variable: average test score in the district.

Regressor	(1)	(2)	(3)	(4)	(5)
Student-teacher ratio (X_1)	-2.28** (0.52)	-1.10* (0.43)	-1.00** (0.27)	-1.31** (0.34)	-1.01** (0.27)
Percent English learners (X_2)		-0.650** (0.031)	-0.122** (0.033)	-0.488** (0.030)	-0.130** (0.036)
Percent eligible for subsidized lunch (X_3)			-0.547** (0.024)		-0.529** (0.038)
Percent on public income assistance (X_4)				-0.790** (0.068)	0.048 (0.059)
Intercept	698.9** (10.4)	686.0** (8.7)	700.2** (5.6)	698.0** (6.9)	700.4** (5.5)
Summary Statistics					
<i>SER</i>	18.58	14.46	9.08	11.65	9.08
$\bar{R}^2$	0.049	0.424	0.773	0.626	0.773
<i>n</i>	420	420	420	420	420

These regressions were estimated using the data on K-8 school districts in California, described in Appendix 4.1. Heteroskedasticity-robust standard errors are given in parentheses under coefficients. The individual coefficient is statistically significant at the *5% level or **1% significance level using a two-sided test.

Testing Single Restrictions Involving Multiple Coefficients

- Sometimes economic theory suggests a single restriction that involves two or more regression coefficients.
- For example, economic theory might suggest that $\beta_1 = \beta_2$. In this case, the task is to test $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$.
- This null hypothesis has only one restriction $q = 1$ but it involves two coefficients so we need to modify the method to test the hypothesis.
- There are two approaches
 - ① **Perform the test directly:** some software, including R or STATA, lets you test restrictions using multiple coefficients directly.
 - ② **Rearrange (“transform”) the regression:** rearrange the regressors so that the restriction becomes a restriction on a single coefficient in an equivalent regression.

Testing Single Restrictions Involving Multiple Coefficients

Performing the Test Directly

$$TestScore_i = \beta_0 + \beta_1 \times STR + \beta_2 \times Expn_i + \beta_3 \times PctEL_i + u_i,$$

$$H_0 : \beta_1 = \beta_2 \text{ vs. } H_1 : \beta_1 \neq \beta_2.$$

Testing Single Restrictions Involving Multiple Coefficients

Rearrange (“transform”) the regression

$$Y_i = \beta_0 + \beta_1 X_{1,i} + \beta_2 X_{2,i} + u_i, \quad (9)$$

$$H_0 : \beta_1 = \beta_2 \text{ vs. } H_1 : \beta_1 \neq \beta_2.$$

Add and subtract $\beta_2 X_{1,i}$ and rearrange to get

$$Y_i = \beta_0 + \gamma_1 X_{1,i} + \beta_2 R_i + u_i, \quad (10)$$

where $\gamma_1 = \beta_1 - \beta_2$ and $R_i = X_{1,i} + X_{2,i}$. Regressions 9 and 10 have the same R^2 , predicted values and residuals.

Testing $\beta_1 = \beta_2$ is equivalent to the usual test $\gamma_1 = 0$.

Confidence Sets for Multiple Coefficients

Recall the meaning of a 95% confidence interval

- The set of points that cannot be rejected at the 5% significance level;
- a set-valued function of the data (an interval) containing the true parameter value 95% of the time in hypothetical repeated samples.
- Example, if our regression output is

$$\widehat{TestScore} = \underset{(10.4)}{698.9} - \underset{(0.52)}{2.28} \times STR,$$

$$R^2 = 0.05, \quad SER = 18.6.$$

the 95% confidence interval for β_1 is $[-3.30, -1.26]$.

- If we tested the null hypothesis $H_0 : \beta_1 = \beta_{1,0}$ vs. $\beta_1 \neq \beta_{1,0}$ using a *t-statistic* at the 5% significance level, we would not reject H_0 for all the values of $\beta_{1,0}$ between -3.3 and -1.25 .

Confidence Sets for Multiple Coefficients

Recall our multiple regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_K X_{Ki} + u_i, \quad \forall i = 1, \dots, n. \quad (11)$$

What is a joint confidence set for β_1 and β_2 ?

- A 95% joint confidence set is: the set of coefficient values that cannot be rejected at the 5% significance level.
- Equivalently, a set-valued function of the data that contains the true coefficient(s) in 95% of hypothetical repeated samples.

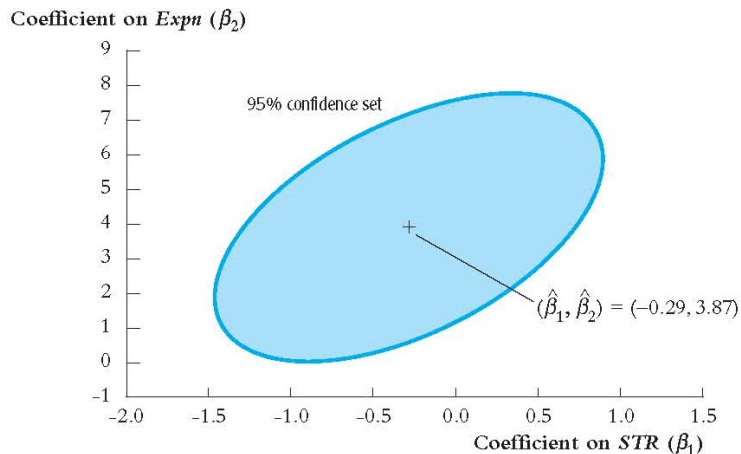
Confidence Sets for Multiple Coefficients

- Imagine you want to construct a 95% confidence set for β_1 and β_2 .
- We learned before how to use the *F-statistic* to test the joint null hypothesis that $H_0 : \beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$ at the 95% confidence level.
- Suppose you were able to test all the possible values of $\beta_{1,0}$ and $\beta_{2,0}$ and save all the pairs $(\beta_{1,0}, \beta_{2,0})$ for which the F-statistic is less than the critical value 3. The set of values above represent the 95% confidence set.
- The procedure described above is good to build intuition but it is impractical because you would need to test for potentially an extremely large number of combinations of values $(\beta_{1,0}, \beta_{2,0})$.

Confidence Sets for Multiple Coefficients

- You can find a 95% confidence set as the set of $(\beta_{1,0}, \beta_{2,0})$ that cannot be rejected at the 5% level using an F-test.
- Let $F(\beta_{1,0}, \beta_{2,0})$ be the (heteroskedasticity-robust) *F-statistic* testing the hypothesis $H_0 : \beta_1 = \beta_{1,0}$ and $\beta_2 = \beta_{2,0}$. The 95% confidence set is $\{(\beta_{1,0}, \beta_{2,0}) : F(\beta_{1,0}, \beta_{2,0}) \leq 3.00\}$.
- 3.00 is the 5% critical value of the $F_{2,\infty}$ distribution.
- This set has coverage rate 95% because the test on which it is based (i.e., the test it “inverts”) has a size of 5%.
- 5% of the time, the test incorrectly rejects the null when the null is true, so 95% of the time it does not; therefore the confidence set constructed as the non-rejected values contains the true values 95% of the time (in 95% of all hypothetical samples).

Confidence Sets for Multiple Coefficients



Model Specification for Multiple Regression

- We want to get an unbiased estimate of the effect on test scores of changing class size, holding constant factors outside the school committee's control –such as outside learning opportunities (museums, etc), parental involvement in education (reading with mom at home?), etc.
- If we could run an experiment, we would randomly assign students (and teachers) to different sized classes. Then STR_i would be independent of all the things that go into u_i and $E[u_i|STR_i] = 0$ and the *OLS* slope estimator in the regression of $TestScore_i$ on STR_i will be an unbiased estimator of the desired causal effect.

Model Specification for Multiple Regression

- But with observational data, u_i depends on additional factors (museums, parental involvement, knowledge of English, etc).
- If you can observe those factors (e.g. $PctEL$), then include them in the regression.
- But usually you can't observe all these omitted causal factors (e.g. parental involvement in homework).
- If additional relevant variables are not observable, sometimes you can include “control variables” which are correlated with these omitted causal factors, but which themselves are not causal.

Model Specification for Multiple Regression

Control Variables

- A control variable W is a variable that is correlated with, and controls for, an omitted causal factor in the regression of Y on X , but which itself does not necessarily have a causal effect on Y .
- Control variables are **NOT** the object of interest; rather they are regressors we include to hold constant factors that, if neglected, could lead to the problem of omitted variable bias.

Model Specification for Multiple Regression

Consider the following regression

$$\widehat{TestScore} = 700.2 - \underset{(5.6)}{1.00} \times STR, - \underset{(0.033)}{0.122} \times PctEL - \underset{(0.024)}{0.547} \times LchPct, \quad (12)$$

$$\overline{R}^2 = 0.773,$$

where $PctEL$ is the percentage of English learners in the school district, and $LchPct$ is the percentage of students receiving a free/subsidized lunch (only students from low-income families are eligible).

Which variable is the variable of interest? Which variables are control variables? Do they have causal components? What do they control for?

Model Specification for Multiple Regression

Answers:

- *STR is the variable of interest.*
- *PctEL* probably has a direct causal effect (school is tougher if you are learning English!). But it is also a control variable: immigrant communities tend to be less affluent and often have fewer outside learning opportunities, and *PctEL* is correlated with those omitted causal variables. *PctEL is both a possible causal variable and a control variable.*
- *LchPct* might have a causal effect (eating lunch helps learning); it also is correlated with and controls for income-related outside learning opportunities. *LchPct is both a possible causal variable and a control variable.*

Model Specification for Multiple Regression

Three interchangeable statements about what makes an effective control variable:

- 1 An effective control variable is one which, when included in the regression, makes the error term uncorrelated with the variable of interest.
- 2 Holding constant the control variable(s), the variable of interest is “as if” randomly assigned.
- 3 Among individuals (entities) with the same value of the control variable(s), the variable of interest is uncorrelated with the omitted determinants of Y .

Model Specification for Multiple Regression

Control variables need not be causal, and their coefficients generally do not have a causal interpretation.

$$\widehat{TestScore} = 700.2 - \underset{(5.6)}{1.00} \times STR, -\underset{(0.033)}{0.122} \times PctEL - \underset{(0.024)}{0.547} \times LchPct,$$

Does the coefficient on $LchPct$ have a causal interpretation?

If so, then we should be able to boost test scores (by a lot!) by simply eliminating the school lunch program so that $LchPct = 0$!

Model Specification for Multiple Regression

Conditional Mean Independence

When control variables are included, the *LSA* (1) $E[u_i | X_{1,i}, \dots, X_{K,i}] = 0$ must not hold.

For example, the coefficient on *LchPct* is correlated with unmeasured determinants of test scores such as outside learning opportunities. But the fact that *LchPct* is correlated with these omitted variables is precisely what makes it a good control variable!

Model Specification for Multiple Regression

Conditional Mean Independence

- If *LSA (1)* does not hold, then what does?
- We need a mathematical statement of what makes an effective control variable. This condition is called **conditional mean independence**: given the control variable, the mean of u_i does not depend on the variable of interest.
- Let X_i denote the variable of interest and W_i denote the control variable(s). W is an effective control variable if conditional mean independence holds: $E[u_i|X_i, W_i] = E[u_i|W_i]$
- If W is a control variable, then conditional mean independence replaces *LSA (1)* –it is the version of *LSA (1)* which is relevant for control variables.

Model Specification for Multiple Regression

Conditional Mean Independence

$$Y = \beta_0 + \beta_1 X + \beta_2 W + u, \quad (13)$$

where X is the variable of interest and W is an effective control variable so that conditional mean independence holds if $E[u_i|X_i, W_i] = E[u_i|W_i]$.

Model Specification for Multiple Regression

Regression With Control Variables

Suppose that in addition to conditional mean independence the Least Square Assumptions (1) and (3) hold, then

- β_1 has a causal interpretation.
- $\hat{\beta}_1$ is unbiased.
- The estimator of the coefficient on the control variable, $\hat{\beta}_2$ is, in general, biased. This is because the control variable is correlated with omitted variables in the error term, so that is subject to omitted variable bias.
- This is not a problem for us because we do not aim to interpret the coefficient of the control variable.

Model Specification in Theory and in Practice

- 1. Identify the variable/s of interest.
- 2. Think of the omitted causal effects that could result in omitted variable bias.
- 3. Include those omitted causal effects if you can or, if you can not, include variables correlated with them that serve as control variables. The control variables are effective if the conditional mean independence assumption plausibly holds (i.e., if u is uncorrelated with the variable/s of interest once the control variables are included). This results in a “base” or “benchmark” model.
- 4. Specify a range of plausible alternative models, which include additional candidate variables.

Model Specification in Theory and in Practice

- 5. Estimate your base model and plausible alternative specifications (“sensitivity checks”).
 - Does a candidate variable change the coefficient of interest (β 's)?
 - Is a candidate variable statistically significant?
 - Use judgement, not a mechanical recipe...
- 6. Do not just try to maximize R^2 ! It is easy to fall into the trap of maximizing the R^2 and $\bar{R}^2$, but this loses sight of our real objective; an unbiased estimator of the parameter of interest. Remember, R^2 and $\bar{R}^2$ are measures of fit, they just tell you whether regressors are good at predicting the values of the dependent variable in the sample, not whether there exist a causal relationship between dependent and independent variables.

Interpreting the R^2 and $\overline{R}^2$

- An increase in R^2 or $\overline{R}^2$ does not necessarily mean that an added variable is statistically significant –this must be determined using hypotheses tests.
- A high R^2 or $\overline{R}^2$ does not mean the regressors are a true cause of the dependent variable.
- A high R^2 or $\overline{R}^2$ does not mean you have eliminated omitted variable bias.
- A high R^2 or $\overline{R}^2$ does not mean that you have the most appropriate set of regressors, nor does a low R^2 or $\overline{R}^2$ necessarily mean that you have an inappropriate set of regressors.

Analysis of the Test Score Data Set

- 1. Identify the variable/s of interest.

Test Scores and STR

- 2. Possible omitted variables.

Whether the students know English; outside learning opportunities; parental involvement; teacher quality (if teacher salary is correlated with district wealth) –there is a long list!

- 3. Include those omitted causal effects if you can or, if you can not, include control variables.

Many of the omitted causal variables are hard to measure, so we need to find control variables. These include PctEL (both a control variable and an omitted causal factor) and measures of district wealth.

Analysis of the Test Score Data Set

- 4. Specify a range of plausible alternative models, which include additional candidate variables.

It is not clear which of the income-related variables will best control for the many omitted causal factors such as outside learning opportunities, so the alternative specifications include regressions with different income variables. The alternative specifications considered here are just a starting point, not the final word!

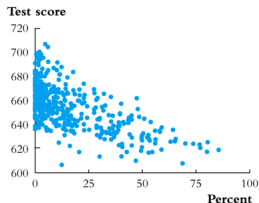
Analysis of the Test Score Data Set

- 5. Estimate your base model and plausible alternative specifications (“sensitivity checks”).

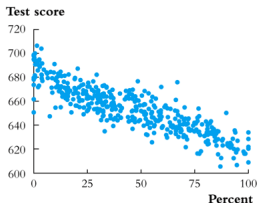
We consider three variables controlling for background characteristics of the students that could affect test scores: the fraction of students who are still learning English, the percentage of students who are eligible for receiving a subsidized or free lunch at school, and the percentage of students in the district whose families qualify for California income assistance program

Case Study

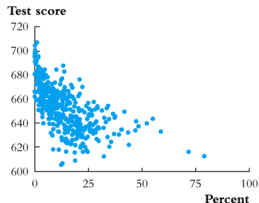
FIGURE 7.2 Scatterplots of Test Scores vs. Three Student Characteristics



(a) Percentage of English language learners



(b) Percentage qualifying for reduced price lunch



(c) Percentage qualifying for income assistance

A Digression on Reporting Results

- Different model specifications lead to a number of regressions that we want to report. It is awkward and difficult to read several regressions written out in equation form (as we did in chapter 5). Instead it is conventional to report them in a table.
- A table of regression results should include
 - Estimated regression coefficients.
 - Standard errors.
 - Measures of fit.
 - Number of observations.
 - Relevant F-statistics, if any.
 - Any other pertinent information.

Case Study

TABLE 7.1 Results of Regressions of Test Scores on the Student–Teacher Ratio and Student Characteristic Control Variables Using California Elementary School Districts

Dependent variable: average test score in the district.

Regressor	(1)	(2)	(3)	(4)	(5)
Student–teacher ratio (X_1)	−2.28** (0.52)	−1.10* (0.43)	−1.00** (0.27)	−1.31** (0.34)	−1.01** (0.27)
Percent English learners (X_2)		−0.650** (0.031)	−0.122** (0.033)	−0.488** (0.030)	−0.130** (0.036)
Percent eligible for subsidized lunch (X_3)			−0.547** (0.024)		−0.529** (0.038)
Percent on public income assistance (X_4)				−0.790** (0.068)	0.048 (0.059)
Intercept	698.9** (10.4)	686.0** (8.7)	700.2** (5.6)	698.0** (6.9)	700.4** (5.5)
Summary Statistics					
<i>SER</i>	18.58	14.46	9.08	11.65	9.08
$\bar{R}^2$	0.049	0.424	0.773	0.626	0.773
<i>n</i>	420	420	420	420	420

These regressions were estimated using the data on K-8 school districts in California, described in Appendix 4.1. Heteroskedasticity-robust standard errors are given in parentheses under coefficients. The individual coefficient is statistically significant at the *5% level or **1% significance level using a two-sided test.